Maths Revision Notes

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Chapter 1

Pure Year I

1.1 Algebraic Expressions

1.2 Quadratics

A quadratic is an expression in the form $ax^2 + bx + c$, for some real constants a, b and c. The solutions for a quadratic in the form $ax^2 + bx + c = 0$ are given by the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{1.1}$$

1.2.1 The Discriminant

From the quadratic formula (equation 1.1), we can see that if $b^2 - 4ac < 0$, the square root will give an imaginary number, meaning that the equation has no real roots. This part of the formula is called the discriminant, written Δ:

$$\Delta = b^2 - 4ac \tag{1.2}$$

Using the discriminant, we can tell whether a quadratic has one distinct real root, two distinct real roots, or no real roots (two complex roots).

- **No real roots:** if $\Delta < 0$, then $\sqrt{\Delta} \notin \mathbb{R} \implies x \notin \mathbb{R}$, so there are no real roots.
- **One distinct real root:** if $\Delta = 0$, then both values of x are equal, so both of the roots are the same ($x = \frac{-b}{2a}$), thus the quadratic has only one distinct real root. The Fundamental Theorem of Algebra says that for an nth degree polynomial, there exists n roots, so the single distinct root is technically two roots with the same value.
- **Two distinct real roots:** if $\Delta > 0$, then the value of $+\sqrt{\Delta}$ is not the same as the value of $-\sqrt{\Delta}$, so the two roots are both real and different (since $a \neq -a \ \forall \ a \in \mathbb{R} \setminus \{0\}$ and $\sqrt{\Delta} \neq 0$ if $\Delta > 0$), giving two distinct real roots.

1.2.2 Factorising

A quadratic $ax^2 + bx + c = 0$ can be factorised into a pair of binomials $a(x - \alpha)(x - \beta) = 0$, where α and β are the roots of the polynomial. To find α and β, find two values m and n that multiply to ac and sum to b. With these values, the expression $ax^2 + bx + c$ can be rearranged into $(ax^2 + mx) + (nx + mn)$. From these brackets, find a common value to factor out of both, and then it will simplify to a pair of binomials.

This can then be used to get the roots. Set each bracket individually to 0, and then solve for α or β.

1.2.3 Quadratic Inequalities

A quadratic inequality is an expression one of these forms:

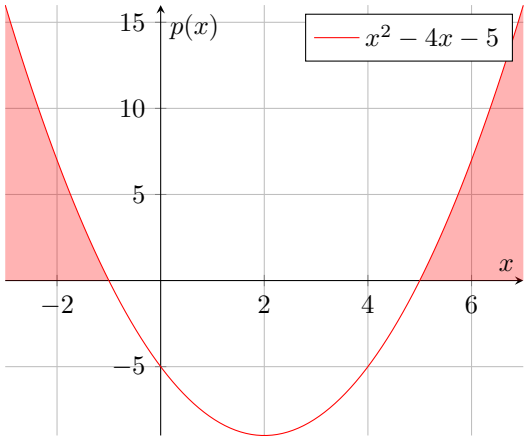
- $ax^2 + bx + c > 0$
- $ax^2 + bx + c < 0$
- $ax^2 + bx + c \geq 0$
- $ax^2 + bx + c \leq 0$

Because the inequality does not set the quadratic equal to a value, but instead allows it to be equal to a range of values, the value of x can also be one in a range.

To solve quadratic inequalities, first find the roots of the quadratic, α and β. Then, look at a graph of the function to find where the value of the quadratic is less than/greater than 0.

For example, let $p(x) = x^2 - 4x - 5$. The roots of $p(x) = 0$ are -1 and 5, so we can then look at a graph to find where $p(x) > 0$, where $p(x) < 0$, where $p(x) \geq 0$, and where $p(x) \leq 0$.

On the graph below, the graph is shaded red where $p(x) > 0$ (where the line is above the x-axis). This means that the values of x are anywhere outside of the interval from -1 to 5.



This means that $x < -1$ or $x > 5$. If the inequality was $p(x) \geq 0$, the solution would alos have $\leq$ and $\geq$ symbols – $x \leq -1$ or $x \geq 5$.

For $p(x) < 0$, we can see that the values of x where $p(x)$ is below the x-axis would be where $-1 < x < 5$. Thus, for $p(x) \leq 0$, $-1 \leq x \leq 5$.

The set of values that satisfy an equation or inequality is called the solution set.

1.2.4 Set-Builder Notation

A set of values can be written using set-builder notation as:

$$\{x \in S : \phi(x)\} \tag{1.3}$$

Where $\phi(x)$ is a condition which can be true or false, based on the value of x, and S is a set containing x. The set represents the set of all values x in the set S that satisfy $\phi(x)$. The '$\in S$' part can be omitted if it is obvious which set is being talked about, usually $S = \mathbb{R}$.

An example of set-builder notation would be the set of all integers greater than 2:

$$\{x \in \mathbb{Z} : x > 2\} \tag{1.4}$$

This would be equivalent to the set:

$$\{3, 4, 5, \cdots\} \tag{1.5}$$

However, the condition $\phi(x)$ can be more complex; ϕ (I have removed the '(x)' for simplicity, but ϕ has the same meaning as $\phi(x)$) can be built from other conditions χ and ψ using boolean logic:

$$\chi \text{ and } \psi: \ \chi \wedge \psi \tag{1.6}$$
$$\chi \text{ or } \psi: \ \chi \vee \psi \tag{1.7}$$
$$\text{not } \psi: \ \neg \psi \tag{1.8}$$

Alternatively, they can be written using unions and intersections of multiple sets:

$$\{x : \chi \wedge \psi\} = \{x : \chi\} \cap \{x : \psi\} \tag{1.9}$$
$$\{x : \chi \vee \psi\} = \{x : \chi\} \cup \{x : \psi\} \tag{1.10}$$
$$\{x \in S : \neg \psi\} = S \setminus \{x : \psi\} \tag{1.11}$$

Set-builder notation can be used to represent solution sets. For example, the set of values that satisfy $\alpha < x < \beta$ can be written:

$$\{x \in \mathbb{R} : \alpha < x < \beta\} \tag{1.12}$$

The set of values that satisfy $x < \alpha$ or $\beta < x$ (where $\alpha < \beta$) can be written:

$$\{x \in \mathbb{R} : (x < \alpha) \vee (\beta < x)\} = \{x \in \mathbb{R} : x < \alpha\} \cup \{x \in \mathbb{R} : \beta < x\} \tag{1.13}$$

Alternatively, this set also represents all values of x in $\mathbb{R}$ that are not in the closed interval $[\alpha, \beta]$, thus could be written $x \in \mathbb{R} \setminus [\alpha, \beta]$.

For a quadratic $p(x) = 0$, the solution set would not require set-builder notation, since there are only one or two values, so it can be written $\{\alpha, \beta\}$. If there are no real roots, the set of real roots is empty, and can be written as the empty set $\emptyset = \{\}$.

1.2.5 Completing the Square

Theorem 1.2.1 (Completing the Square)**.** *A quadratic in the form ax^2+bx+c can be written in the form $a(x+m)^2+n$, where $m=\frac{b}{2a}$ and $n=-\frac{b^2}{4a}+c$:*

$$a^2+bx+c=a\left(x+\frac{b}{2a}\right)^2-\frac{b^2}{4a}+c \tag{1.14}$$

Proof. We will begin by factoring a out of the first two terms of the original quadratic to get:

$$ax^2+bx+c=a\left(x^2+\frac{b}{a}\right)+c \tag{1.15}$$

Now, the $\left(x^2+\frac{b}{a}\right)$ part can almost be factorised into $\left(x+\frac{b}{2a}\right)^2$, however expanding that will give an extra $\left(\frac{b}{2a}\right)^2$ term. This means that it can be rewritten as:

$$\left(x^2+\frac{b}{a}\right)=\left(x+\frac{b}{2a}\right)^2-\frac{b^2}{4a^2} \tag{1.16}$$

This can be put back into the formula to get:

$$ax^2+b+c=a\left(\left(x+\frac{b}{2a}\right)^2-\frac{b^2}{4a^2}\right)+c \tag{1.17}$$

Moving the $\frac{b^2}{4a^2}$ term out of the brackets (and thus multiplying by a):

$$ax^2+bx+c=a\left(x+\frac{b}{2a}\right)^2-\frac{b^2}{4a}+c \tag{1.18}$$

□

1.2.6 Quadratic Turning Points

A quadratic $p(x)$ is a parabola, so has a turning point where $\frac{dp}{dx}=0$. This can be found in one of two ways:

- **Completing the Square:** use completing the square (see section 1.2.5) to put the quadratic in the form $a(x-m)^2+n$. This is the same as using graph transformations to move the parabola ax^2 (which has a turning point $(0,0)$) right by m (subtract m from the x value), and then moving it up by n (add n to the overall value). This means that the turning point is going to be $(0,0)$, moved right by m and up by n, which is (m,n). Substituting in the values for the quadratic ax^2+bx+c, the turning point is $\left(-\frac{b}{2a},-\frac{b^2}{4a}+c\right)$.
- **Differentiation:** since the turning point is the point where $\frac{dp}{dx}=0$, it can be found by finding the derivative of $p(x)$, which (for $p(x)=ax^2+bx+c$) is $2ax+b$. Then, set $2ax+b$ to 0, giving $x=\frac{b}{2a}$. This can then be substituted into $p(x)$ to give the y-coordinate ($y=p\left(\frac{b}{2a}\right)$).

1.3 Equations and Inequalities

1.4 Graphs and Transformations

1.5 Straight Line Graphs

1.6 Circles

1.7 Algebraic Methods I

1.8 The Binomial Expansions

1.9 Trigonometric Ratios

1.9.1 The Sine Rule

Theorem 1.9.1 (Sine Rule)**.** *The sine rule states that, for a triangle with sides a, b, c, and angles A, B, C opposite to the corresponding side (A opposite to a, etc.), the following holds:*

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \tag{1.19}$$

Proof. We will begin by drawing the triangle described in theorem 1.9.1, with a line with length h from C to AB, meeting AB at a right-angle:

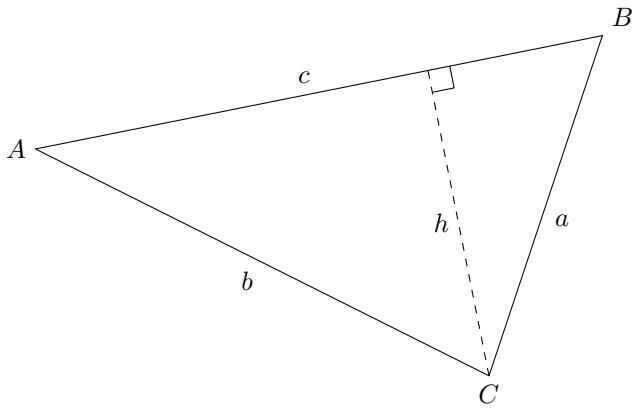


Figure 1.1: A triangle with sides a, b, c, and angles A, B and C

From this, we can see that $\sin A = \frac{h}{b}$, using the rules of right-angled trigonometry ($\sin\theta = \frac{O}{H}$). Thus, $b\sin A = h$. Using the same logic for the other right-angled triangle, we can also derive $a\sin B = h$. We can now set the two expressions equal to each other to find:

$$b\sin A = a\sin B$$
$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

We can now use another line from B to AC to do the same process with angles A and C to prove that $\frac{a}{\sin A} = \frac{c}{\sin C}$, and thus:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \tag{1.20}$$

□

The sine rule can be used to find missing sides by rearranging for a, b or c, or all of the fractions can be flipped to put the sine rule in the form:

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} \tag{1.21}$$

Which can then be further rearranged to find a angle, e.g.:

$$A = \arcsin\left(\frac{a \sin B}{b}\right) \tag{1.22}$$

1.9.2 The Cosine Rule

Theorem 1.9.2 (Cosine Rule)**.** *The cosine rule states the, for a triangle with sides a, b, c, and angles A, B, C opposite to the corresponding side (A oppsite to a, etc.), the following holds:*

$$a^2 = b^2 + c^2 - 2bc \cos A \tag{1.23}$$

Proof. We will begin by drawing the triangle described in 1.9.2, with a line with length h from C to AB, meeting AB at a right-angle, and splitting it into length x and $c - x$:

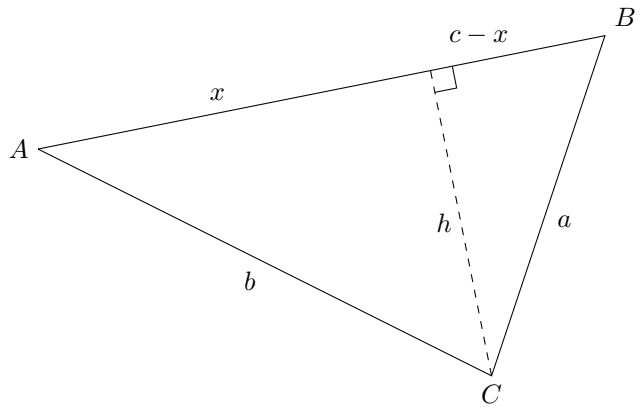


Figure 1.2: A triangle with sides a, b, c, and angles A, B and C

From here, we can use the Pythagorean theorem to see that $b^2 - x^2 = h^2$ and that $a^2 - (c - x)^2 = h^2$. Setting them equal to each other and rearranging, we get:

$$\begin{aligned} b^2 - x^2 &= a^2 - (c - x)^2 \\ b^2 - x^2 &= a^2 - c^2 + 2cx - x^2 \\ b^2 &= a^2 - c^2 + 2cx \end{aligned}$$

Using the rules of right-angled trigonometry ($\cos\theta = \frac{A}{H}$), we can see that $\cos A = \frac{x}{b}$ thus $x = b\cos A$, which we can substitute into $b^2 = a^2 - c^2 + 2cx$ to get:

$$b^2 = a^2 - c^2 + bc\cos A \tag{1.24}$$

Which can be rearranged into:

$$a^2 = b^2 + c^2 - 2bc\cos A \tag{1.25}$$

□

1.9.3 The Area of a Triangle

1.9.4 Trigonometric Graphs

1.9.5 Trigonometric Graph Transformations

1.10 Trigonometric Equations and Identities

1.11 Vectors I

1.12 Differentiation I

1.13 Integration I

1.14 Exponentials and Logarithms

Chapter 2

Pure Year II

2.1 Algebraic Methods II

2.2 Functions and Graphs

2.3 Sequences and Series

2.4 Binomial Expansion

2.5 Radians

2.6 Trigonometric Functions

2.7 Trigonometry and Modelling

2.8 Parametric Equations

2.9 Differentiation II

2.10 Numerical Methods

2.11 Integration II

2.12 Vectors II

Chapter 3

Mechanics

3.1 Mechanics Year I

3.1.1 Modelling in Mechanics

3.1.2 Constant Acceleration

When modelling objects with constant acceleration, (i.e. objects with a constant force being applied and with constant mass), there is a set of equations that relate the object's displacement, initial velocity, final velocity, acceleration, and the time period over which its velocity changes. These equations are called the SUVAT equations.

The first equation is $s = \frac{1}{2}(u + v)t$.

Proof. Drawing a velocity-time graph of an object gives a linear function, since $\frac{dv}{dt} = a$, which is constant, so the slope must be constant. Thus, the area under it (s, since $s = \int v\,dt$) is a trapezium. Given that the area of a trapezium is $A = \frac{1}{2}(a + b)h$, where a and b are the lengths of the two sides and h is the distance between them, we can find that s is $\frac{1}{2}(u+v)t$, since the two sides (the left and right of the graph, at time 0 and t) are u and v, since the velocity at time 0 is the initial velocity u and the velocity at time t is the final velocity v. The distance between the two velocities is t, thus giving the final formula $s = \frac{1}{2}(u + v)t$. □

The second formula is $v = at + u$.

Proof. Given that:

$$v = \int a\,dt \tag{3.1}$$

We can integrate a (given that a is constant) to get:

$$v = at + C$$

Here, C is the value of v at $t = 0$, which is the initial velocity u, thus:

$$v = at + u \tag{3.2}$$

□

From these two equations, the other 3 can be derived.

- $s = ut + \frac{1}{2}at^2$

  Proof. We will substitute the equation $v = at + u$ into $s = \frac{1}{2}(u+v)t$. This gives:

  $$\begin{aligned} s &= \frac{1}{2}(u + at + u)t \\ &= \frac{1}{2}(2u + at)t \\ &= ut + \frac{1}{2}at^2 \end{aligned}$$

  □

- $s = vt - \frac{1}{2}at^2$

Proof. We will rearrange $v = at + u$ to $u = v - at$ and substitute it into $s = \frac{1}{2}(u + v)t$ to get:

$$\begin{aligned} s &= \frac{1}{2}(v - at + v)t \\ &= \frac{1}{2}(2v - at)t \\ &= v - \frac{1}{2}at^2 \end{aligned}$$

□

- $v^2 = u^2 + 2as$

Proof. We will first rearrange $s = \frac{1}{2}(u+v)t$ into $s = \left(\frac{u+v}{2}\right) t$ and $v = u+at$ into $t = \left(\frac{v-u}{a}\right)$. Substituting the second one into the first gives:

$$s = \left(\frac{u+v}{2}\right)\left(\frac{v-u}{a}\right) \tag{3.3}$$

This creates the factored form of a difference of two squares in the numerator, which simplifies to $v^2 - u^2$. The denominators multiply to get $2a$. Rearranging then gives:

$$\begin{aligned} s &= \frac{v^2 - u^2}{2a} \\ 2as &= v^2 - u^2 \\ v^2 &= u^2 + 2as \end{aligned}$$

□

If an object is in free-fall, its motion can be modelled with constant acceleration, taking its acceleration to be $g = 9.8\,\mathrm{m\,s^{-2}}$ downwards, which is (approximately) the acceleration due to gravity on Earth at sea-level.

3.1.3 Forces and Motion

3.1.4 Variable Acceleration

3.2 Mechanics Year II

3.2.1 Moments

3.2.2 Forces and Friction

3.2.3 Projectiles

3.2.4 Applications of Forces

3.2.5 Further Kinematics

Chapter 4

Statistics

4.1 Statistics Year I

4.1.1 Data Collection

4.1.2 Measures of Location and Spread

4.1.3 Representations of Data

4.1.4 Correlation

4.1.5 Probability

4.1.6 Statistical Distributions

4.1.7 Hypothesis Testing

4.2 Statistics Year II

4.2.1 Regression, Correlation and Hypothesis Testing

4.2.2 Conditional Probability

4.2.3 The Normal Distribution

Appendix A

Appendix: Formula Sheet

Appendix B

Appendix: Non-Formula Sheet